

SE LAB-01

Local Sports Tournament & Bracket Manager

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Section:- K

Functional requirements-(F001 to F005)

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall automatically generate seeded single/double elimination tournament brackets based on initial team rankings.	High	Pass: Winner of Match 1 advances automatically to Match 2 slot; Fail: Bracket produces invalid match pairings.	Accurate team registration and rankings are required before generating a valid tournament bracket.
FR-002	Functional	The system shall automatically generate seeded single-elimination or double-elimination tournament brackets based on the registered teams and their rankings.	High	Pass: A valid bracket is generated with teams correctly seeded and no duplicate match pairings. Fail: The system produces missing, duplicate, or invalid match pairings.	Automated bracket generation reduces manual errors and ensures a valid tournament structure.
FR-003	Functional	The system shall allow authorized users to record match scores and determine	High	Pass: A submitted score is stored and the winner is correctly	Accurate score recording and winner

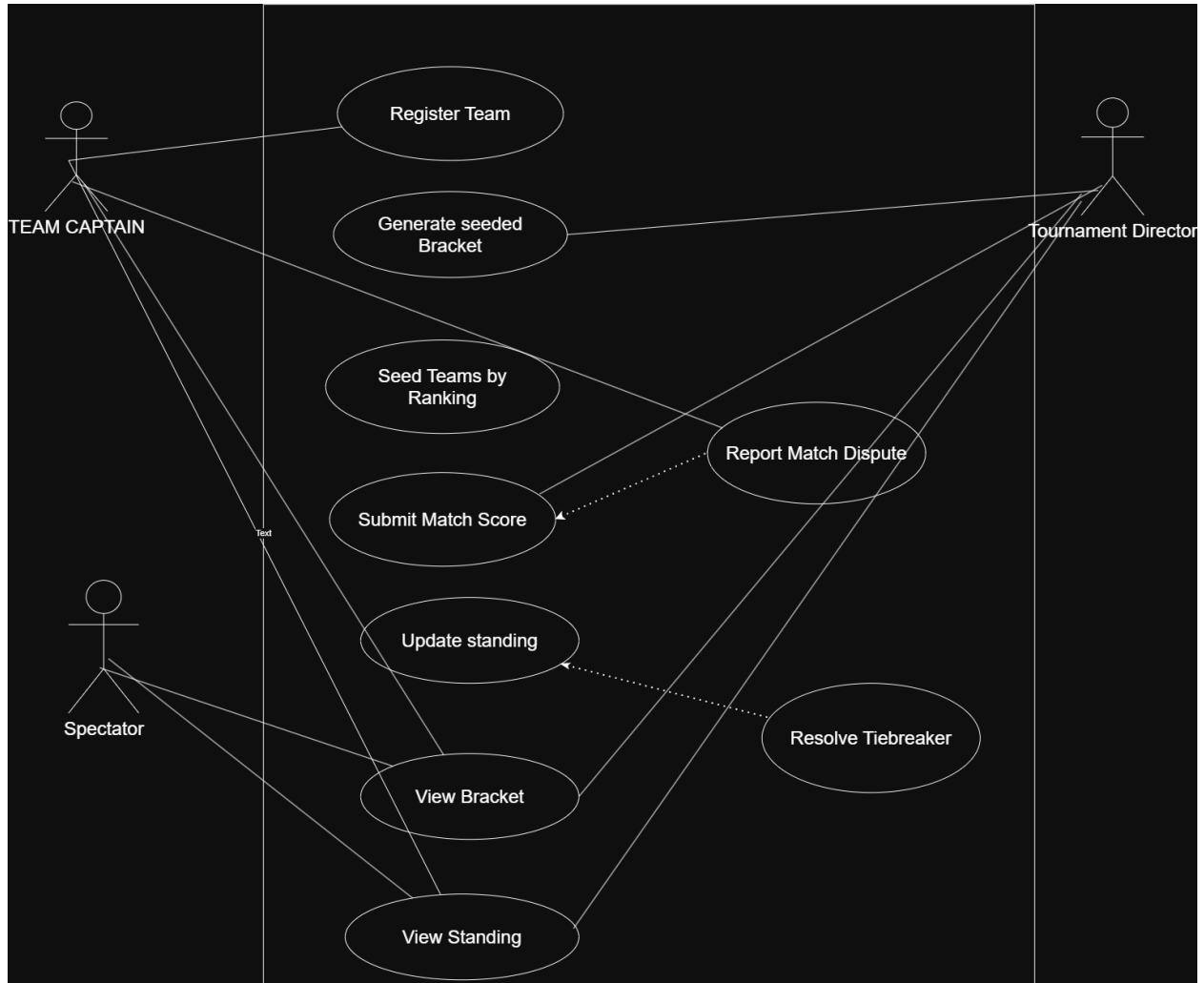
ID	Type	Description	Priority	Acceptance Criteria	Rationale
		the winner, including applying tiebreaker rules when required.		determined. Fail: Invalid scores are rejected or an incorrect winner is calculated.	determination are essential for progressing the tournament.
FR-004	Functional	The system shall automatically advance the winner of a completed match to the appropriate next match in the tournament bracket.	High	Pass: The winner appears in the correct next-match slot. Fail: The winner is placed in an incorrect or unavailable slot.	Automatic progression keeps the tournament bracket consistent after every completed match.
FR-005	Functional	The system shall recalculate and publish updated league standings and tournament leaderboards after valid match results are recorded.	High	Pass: Standings, goal differentials, and leaderboard information reflect the latest valid result. Fail: Published standings contain outdated or inconsistent results.	Participants need current standings to track tournament progress and rankings.

Non-Functional requirements-(NFR001 to NFR002)

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Security]	Standings, goal differentials, and tournament leaderboard updates must recalculate within 200 ms of score submission.	High	Pass: After a valid score is submitted, the system recalculates and displays the updated standings, goal differentials, and tournament leaderboard within	Fast updates allow participants and tournament administrators to see current results without noticeable delay.

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				200 ms under the specified test load. Fail: Any of the required updates take more than 200 ms to recalculate after score submission.	
NFR-002	Security	The system shall restrict tournament-management operations such as team management, bracket generation, and result administration to authorized Tournament Directors.	High	Pass: An authorized Tournament Director can perform restricted operations. Fail: An unauthorized user attempting a restricted operation is denied access.	Prevents unauthorized users from modifying tournament data and maintains the integrity of tournament results.

UML DIAGRAM:-



Actor	Actor Description
Team Captain	Registers their team, views the bracket and live standings, and can flag a disputed match score.

Actor	Actor Description
Tournament Director	Administers the tournament — generates the bracket, submits match results, and resolves tiebreakers.
Spectator	A public/community viewer with read-only access to the bracket and standings (no login-gated actions). Added beyond the two stakeholders named in the problem statement, since a public tournament platform in the "Media, Events & Community" category implies a viewer role — flag this with your TA if only the two listed actors are expected.

Use Cases

ID	Use Case	Primary Actor(s)
UC-01	Register Team	Team Captain
UC-02	Generate Seeded Bracket	Tournament Director
UC-03	Submit Match Score	Tournament Director
UC-04	View Bracket	Team Captain, Tournament Director, Spectator
UC-05	View Standings	Team Captain, Tournament Director, Spectator
UC-06	Seed Teams by Ranking («include» of UC-02)	Tournament Director
UC-07	Update Standings («include» of UC-03)	Tournament Director
UC-08	Resolve Tiebreaker («extend» of UC-07)	Tournament Director
UC-09	Report Match Dispute («extend» of UC-03)	Team Captain

Use-Case Flow: UC-03 — Submit Match Score

Actor: Tournament Director

Preconditions:

- The Tournament Director is authenticated and has administrative access to the active tournament.
- The match exists in the current bracket round, with both competing teams assigned.
- The match has not already been marked as completed.

Postconditions:

- The match score is recorded and the match is marked complete.
- Standings and goal differentials are recalculated (via UC-07, «include»).
- The winning team automatically advances to its next bracket slot (for elimination matches).

Main Success Scenario:

1. Tournament Director selects the match to update from the active bracket view.
2. System displays the match entry form (team names, current round).
3. Tournament Director enters the final score for each team.
4. System validates the score (non-negative integers, valid match state).
5. System records the score and marks the match as complete.

6. System triggers UC-07 (Update Standings), recalculating the leaderboard within the 200 ms latency target (NFR-001).
7. System advances the winning team to its next bracket slot and updates the bracket view.
8. System confirms the update to the Tournament Director.

Alternate Flow — A1: Score Results in a Tie Requiring Tiebreaker

Trigger: at step 6, the recalculated standings show two or more teams tied on the primary ranking criteria

1. System detects the tie condition during standings recalculation.
2. System invokes UC-08 (Resolve Tiebreaker), applying the configured tiebreak rule (e.g., goal differential, then head-to-head result).
3. System recalculates the standings order using the tiebreak outcome.
4. Flow resumes at main scenario step 7 (bracket advancement).

Alternate Flow — A2: Invalid Score Entry Trigger: at step 4, validation fails .

1. System rejects the submission and displays a validation error.
2. Tournament Director corrects the input.
3. Flow resumes at main scenario step 3

